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AMIT KATIYAR
(MCA-JNU)



0512-3163515



9935985550



ENGLISH

1. Select the correct word from the options.

After you _____, your lifestyle usually changes.

- (a) play (b) Sleep
(c) go shopping (d) get married

English – Sentence Completion – 2021

2. Select the most suitable synonym:

SWANKY

- (a) posh (b) poor
(c) plain (d) modest

English – Synonyms – 2021

3. Which is the nearest meaning of the phrase **to rain cats and dogs**?

- (a) heavy rain
(b) to rain non stop
(c) fight among cats and dogs
(d) drizzling

English – Idioms & Phrases – 2021

4. Fill in the blanks with the right word:

_____ he is well known, he is very honest

- (a) Despite the fact that
(b) In spite of the fact that
(c) Not with standing
(d) Even if

English – Conjunctions / Grammar – 2021

5. Select the most suitable antonym:

VAGABOND

- (a) unsettled (b) fly-by-night
(c) aimless (d) permanent

English – Antonyms – 2021

6. Choose the correct spelling

- (a) opportunity (b) oportunity
(c) ooportunity (d) aportunity

English – Spellings – 2021

7. Identify the meaning of the following idiom from the options given.

Spill the beans

- (a) To be untidy
(b) to be very talkative
(c) to reveal something that is supposed to be kept a secret
(d) To leave a place quickly

English – Idiom Meaning – 2021

8. Which is the nearest meaning of the phrase **out of blue**

- (a) stranger (b) unexpectedly
(c) to bring happiness (d) emerging out of blue colour.

English – Phrase Meaning – 2021

9. Fill in the blank with the right option.

A treasurer mustthe money he spends.

- (a) allow for (b) give account for
(c) account for (d) As for

English – Verb Usage – 2021

10. The phrase, A hard nut to crack means

- (a) an interesting problem (b) a difficult problem
(c) a unique problem (d) a simple problem

English – Idioms – 2021

REASONING

11. Choose the number which is different from others in the group.

- (a) 8 (b) 64
(c) 125 (d) 28

Reasoning – Number Reasoning (Odd One Out) – 2021

12. In a row of trees, one tree is fifth from either end of the row. How many trees are there in the row?

- (a) 8 (b) 9
(c) 10 (d) 11

Reasoning – Ranking / Position – 2021

13. If – means $\times$, $\times$ means $+$, $+$ means $\div$ and $\div$ means $-$, then $40 \times 12 + 3 - 6 \div 60 = ?$

- (a) 7.95 (b) 16
(b) 44 (d) None of these

Reasoning – Symbolic Logic – 2021

14. Choose the correct answer

$39798 + 3798 + 378 = ?$

- (a) 43576 (b) 43974
(c) 43984 (d) 49532

Reasoning – Arithmetic Reasoning – 2021

15. Find the lowest common multiple of 24, 36 and 40.

- (a) 120 (b) 240
(c) 360 (d) 480

Reasoning – LCM & HCF – 2021

16. Find the number which when multiplied by 15 increased by 196.

- (a) 14 (b) 20
(c) 26 (d) 28

Reasoning – Arithmetic Equation – 2021

17. Choose the correct alternative based on relationship

Planet : Orbit :: Projectile : ?

- (a) Trajectory (b) Track
(c) Milky Way (d) Path

Reasoning – Analogy – 2021

18. Choose the word which is least like the other words in the group

- (a) Pistol (b) Sword
(c) Gun (d) Rifle

Reasoning – Odd One Out – 2021

19. Choose the number which is different from others in the group.

- (a) 21 (b) 36 (c) 49 (d) 56

Reasoning – Number Reasoning – 2021

20. Find out the missing number in the following sequence:

1, 3, 3, 6, 7, 9, ?, 12, 21

- (a) 10 (b) 11 (c) 12 (d) 13

Reasoning – Number Series – 2021



21. If in a certain code TWENTY is written as 863985 and ELEVEN is written 323039, how is TWELVE written in that code?

(a) 863203 (b) 863584
(c) 863903 (d) 863063

Reasoning – Coding–Decoding – 2021

22. DGCA stands for

(a) Director General of Civil Association
(b) Direct General of Civil Aviation
(c) Director General of Coast Accounts
(d) Director General of Coal Authority

GK – Abbreviation – 2021

23. The 46th President of the USA is

(a) Donald Trump (b) Joe Biden
(c) Kamala Harris (d) Mike Pence

GK – Current Affairs – 2021

24. According to the NEP 2020, the school system will change to

(a) 5+2+2+4 (b) 5+3+3+4
(c) 5+2+3+4 (d) 5+4+3+4

GK – Education Policy (NEP) – 2021

25. Russian vaccine for COVID-19 is

(a) Sputnik V (b) Sputnik VI
(c) Sputnik (d) Sputnik IV

GK – Science / Health – 2021

26. The difference between two sets {1,2,3} and {1,2,5} is the set

(a) {1} (b) {5}
(c) {3} (d) {2}

Mathematics – Set Theory (Difference of Sets) – 2021

27. The set of positive integers is

(a) infinite (b) finite
(c) subset (d) empty

Mathematics – Set Theory – 2021

28. Which of the following Venn diagram represents the relation between Currency, Rupee and Dollar?



(a) a (b) b (c) c (d) d

Mathematics – Venn Diagram – 2021

29. Let $n(x)$ denotes the number of elements in set X. If $n(A) = p$ and $n(B) = q$, then how many ordered pairs (a,b) are there with $a \in A$ and $b \in B$?

(a) p^2 (b) $p \times q$
(c) $p + q$ (d) $2pq$

Mathematics – Cartesian Product – 2021

30. The number of elements in the power set $P(S)$ of the set $S = \{\{\Phi\}, 1, \{2, 3\}\}$ is

(a) 2 (b) 4
(c) 6 (d) 8

Mathematics – Power Set – 2021

31. A set which contains a definite number of elements is called

(a) proper subset (b) universal set
(c) finite set (d) unit set

Mathematics – Types of Sets – 2021

32. Determine the characteristics of the relation aRb , if $a^2 = b^2$

(a) Transitive and symmetric
(b) Reflexive and asymmetric
(c) Trichotomy, antisymmetric and irreflexive
(d) Symmetric, Reflexive and Transitive

Mathematics – Relations (Properties) – 2021

33. In a room containing 28 people, there are 18 people who speak English, 15 people who speak Hindi and 22 people who speak Kannada, 9 people speak both English and Hindi, 11 people speak both Hindi and Kannada whereas 13 people speak both Kannada and English. How many people speak all the three languages?

(a) 6 (b) 7 (c) 8 (d) 9

Mathematics – Venn Diagram (3 Sets) – 2021

34. Which of the following commands is used to delete a table in SQL?

(a) Delete (b) Drop (c) Truncate (d) Remove

Computer – DBMS (SQL Command) – 2021

35. Which of the following standard algorithms is not a greedy algorithm?

(a) Prim's algorithm (b) Kruskal algorithm
(c) Dijkstra's shortest path algorithm
(d) Bellman-Ford shortest path algorithm

Computer – Algorithms (Greedy) – 2021

36. The waterfall model of software development is

(a) The best model to use for moderate size projects
(b) the best model to use for projects with high risk of requirements change
(c) a reasonable model when requirements are well defined.
(d) a good approach when a working program is required quickly.

Computer – Software Engineering (Waterfall Model) – 2021

37. A graph is a collection of

(a) rows and columns (b) vertices and edges
(c) equations (d) matches

Computer – Graph Theory – 2021

38. Normals distribution is also known as

(a) Cauchy's distribution (b) Laplacian distribution
(c) Gaussian distribution (d) Lagrangian distribution

Mathematics – Probability Distribution – 2021

39. How many distinguishable permutations of the letters are there in the word BANANA?

(a) 720 (b) 120 (c) 60 (d) 360

Mathematics – Permutations – 2021

40. The probability of getting a job for a person is $x/3$. If the probability of not getting the job is $2/3$, then the value of x is

(a) 2 (b) 1 (c) 3 (d) 1.5

Mathematics – Probability – 2021

41. If the standard deviation of x, y and z is p, then the standard deviation of $3x + 5$, $3y + 5$, $3z + 5$ is

(a) $3p + 5$ (b) $3p$
(c) $p + 5$ (d) $9p + 15$

Mathematics – Standard Deviation – 2021



42. A man has 6 friends. In how many ways he can invite one or more of his friends at a dinner?

- (a) 61 (b) 62
(c) 63 (d) 64

Mathematics – Combinations – 2021

43. Find the value of a_4 for the recurrence relation $a_n = 2a_{n-1} + 3$, with $a_0 = 6$.

- (a) 320 (b) 221
(c) 141 (d) 65

Mathematics – Recurrence Relation – 2021

44. Which of the following scheduling algorithms gives minimum average waiting time?

- (a) shortest job first (b) Round robin
(c) priority (d) First come first serve

Computer – CPU Scheduling – 2021

45. If for a square matrix A , $A^2 = A$, then such a matrix is known as

- (a) idempotent matrix (b) orthogonal matrix
(c) null matrix (d) None of the above

Mathematics – Matrices (Idempotent Matrix) – 2021

46. According to the principle of mathematical induction. If $P(k+1) = m^{(k+1)} + 5$ is true, then _____ must be true?

- (a) $P(k) = 3m^{(k)}$ (b) $P(k) = m^{(k)} + 5$
(c) $P(k) = m^{(k+2)} + 5$ (d) $P(k) = m^{(k)}$

Mathematics – Mathematical Induction – 2021

47. The network topology supports bidirectional links between each possible node is

- (a) mesh (b) star (c) ring (d) tree

Computer – Network Topology – 2021

48. If $A = \begin{bmatrix} a & b \\ b & a \end{bmatrix}$ and $A^2 = \begin{bmatrix} \alpha & \beta \\ \beta & \alpha \end{bmatrix}$ then

- (a) $\alpha = 2ab, \beta = 2ab$ (b) $\alpha = a^2 + b^2, \beta = 2ab$
(c) $\alpha = a^2 + b^2, \beta = a^2 - b^2$ (d) $\alpha = 2ab, \beta = a^2 + b^2$

Mathematics – Matrix Multiplication – 2021

49. A CPU-bound program might have

- (a) a few long CPU bursts (b) a few short CPU bursts
(c) many short CPU bursts (d) None of the above

Computer – Operating System (CPU Bound) – 2021

50. Let us consider a square matrix, A of order n with eigen values of a, b, c . Then the eigen values of the matrix A^T could be

- (a) a, b, c (b) $-a, -b, -c$
(c) $a-b, b-c, c-a$ (d) a^{-1}, b^{-1}, c^{-1}

Mathematics – Eigen Values – 2021

51. Which of the following operations is not $O(1)$ for an array of sorted data? You may assume that array elements are distinct

- (a) Find the i th largest element
(b) Delete an element
(c) Find the i th smallest element
(d) All of the above

Computer – Time Complexity (Array Operations) – 2021

52. Informal high-level description of an algorithm in English is called

- (a) function (b) class
(c) Pseudo code (d) None of the above

Computer – Pseudocode – 2021

53. To evaluate an expression without any embedded function calls

- (a) one stack is enough
(b) two stacks are needed
(c) as many stacks as the height of the expression tree are needed
(d) a Turing machine is needed in the general case

Computer – Stack – 2021

54. Which of these operators is used to allocate memory to array variable in Java?

- (a) Malloc (b) Alloc
(c) New (d) New malloc

Computer – Java (Memory Allocation) – 2021

55. What is the worst-case time complexity of linear search algorithm?

- (a) $O(n)$ (b) $O(i)$
(c) $O(\log n)$ (d) $O(n^2)$

Computer – Time Complexity (Linear Search) – 2021

56. Consider the following function implemented in C:

```
void test_fun(int x, int y){
    int *p;
    x=0;
    p = &x;
    y = *p;
    *p = 1;
    printf("%d,%d", x, y)
}
```

The output of invoking test_fun (10, 10) is

- (a) 0, 0 (b) 1, 0 (c) 0, 1 (d) 1, 1

Computer – C Pointers – 2021

57. Which of the following data structures is a linear type?

- (a) Graph (b) Tree
(c) Binary (d) Queue

Computer – Data Structures (Linear) – 2021

58. The smallest integer that can be represented by an 8-bit number in 2's complement form is

- (a) -256 (b) -128 (c) -127 (d) 0

Computer – Number System (2's Complement) – 2021

59. Which among the following best describes polymorphism?

- (a) it is the ability for a message/data to be processed in more than one form
(b) It is the ability for a message/data to be processed in only one form
(c) It is the ability for many messages/data to be processed in one way
(d) It is the ability for undefined message/data to be processed in at least one way

Computer – OOPs (Polymorphism) – 2021

60. Which among the following is the language which supports classes but not polymorphism?

- (a) SmallTalk (b) Ada
(c) Java (d) C++

Computer – Programming Languages – 2021



61. _____ is a template for entities that have common behaviour

- (a) class (b) object
(c) Data (d) Polymorphism

Computer – OOPs (Class) – 2021

62. Which of the following codes is used in Java to represent characters?

- (a) ASCII code (b) Unicode
(c) Byte code (d) None of the above

Computer – Character Encoding (Unicode) – 2021

63. Which operator is having the highest precedence?

- (a) Postfix (b) Unary
(c) Shift (d) Equality

Computer – Operator Precedence – 2021

64. The given numbers are inserted into an empty binary search tree in the given order: 10, 1, 3, 5, 15, 12 and 16. What is the height of the binary search tree.

- (a) 3 (b) 4
(c) 5 (d) 6

Computer – Binary Search Tree – 2021

65. Quicksort is a

- (a) greedy algorithm
(b) divide and conquer algorithm
(c) dynamic programming algorithm
(d) backtracking algorithm

Computer – Sorting Algorithms (Quick Sort) – 2021

66. What happens if the following program is executed in C and C++?

```
#include <stdio.h>
int main(void)
{
    int new = 1;
    printf ("%d", new);
}
```

- (a) Error in both C and C++
(b) A successful run in both C and C++
(c) Error in C and successful execution in C++
(d) Error in C++ and successful execution in C

Computer – C vs C++ Keywords – 2021

67. Which of the following is not a member of class?

- (a) static function
(b) Friend function
(c) const function
(d) Virtual function

Computer – OOPs (Class Members) – 2021

68. Which of the following is not an application of binary search?

- (a) To find the lower/upper bound in an ordered sequence
(b) union of intervals
(c) Debugging
(d) To search in unordered list

Computer – Binary Search Applications – 2021

69. What is the output of this C code?

```
#include<stdio.h>
void main()
{
    int x = 18;
    void *ptr = &x;
    printf ("%f/n", *(float)ptr);
}
```

- (a) compile time error (b) undefined behaviour
(c) 10 (d) 0.000000

Computer – C Type Casting – 2021

70. Prim's algorithm is a/an

- (a) divide and conquer algorithm
(b) greedy algorithm
(c) dynamic programming
(d) approximation algorithm

Computer – Graph Algorithms (Prim's) – 2021

71. Which of the following ways can be used to represent a graph?

- (a) only Adjacency list (b) only adjacency Matrix
(c) only incidence Matrix (d) All of the above

Computer – Graph Representation – 2021

72. In a B* tree, If the search-key value is 8 bytes long, the block size is 512 bytes and the block pointer size is 2 bytes, then maximum order of the B* tree is

- (a) 50 (b) 51 (c) 52 (d) 54

Computer – B-Tree / B*-Tree – 2021

73. The minimum number of functions should be present in a C++ program for its execution is

- (a) 0 (b) 1 (c) 2 (d) 3

Computer – C++ Program Structure – 2021

74. Given an array A = {55, 66, 77, 88, 99} and a key = 88. How many iterations are done until the element is found?

- (a) 1 (b) 3 (c) 4 (d) 2

Computer – Searching (Binary Search) – 2021

75. When one object reference variable is assigned to another object reference variable, then

- (a) a copy of the object is created
(b) a copy of the reference is not created
(c) it is illegal to assign one object reference variable to another object reference variable
(d) a copy of the reference is created

Computer – Java Object Reference – 2021

76. The number of different binary trees with 6 nodes is

- (a) 6 (b) 42
(c) 132 (d) 256

Computer – Binary Trees (Counting) – 2021

77. What is the scope of the variable declared in the user-defined function?

- (a) in whole program (b) only inside the {} block
(c) Both 1 and 2 (d) None of these

Computer – Scope of Variables – 2021



78. Which of the following is a valid class declaration?

- (a) class A {int x;} (b) class B { }
(c) public class A { } (d) object A {int x};

Computer – Java Class Declaration – 2021

79. The time factor when determining the efficiency of an algorithm is measured by

- (a) counting microseconds
(b) counting the number of key operations
(c) counting the number of statements
(d) counting the kilobytes of algorithm

Computer – Algorithm Analysis – 2021

80. The geometrical figure shown below in flowchart represents



- (a) input (b) connector
(c) decision (d) looping

Computer – Flowchart Symbols – 2021

81. The complexity of binary search algorithm is

- (a) $O(n)$ (b) $O(\log n)$
(c) $O(n^2)$ (d) $O(n \log n)$

Computer – Binary Search Time Complexity – 2021

82. In the expression $Y + X'$, Y , which operator will be evaluated first?

- (a) ' (b) + (c) . (d) ,

Computer – Boolean Algebra (Operator Precedence) – 2021

83. Which of the following is false?

- (a) $x + y = y + x$ (b) $x \cdot y = y \cdot x$
(c) $x \cdot x' = 1$ (d) $x + x' = 1$

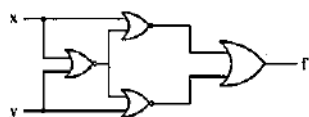
Computer – Boolean Algebra Laws – 2021

84. A Boolean function $x'y' + xy + x'y$ is equivalent to

- (a) $x' + y'$ (b) $x + y$
(c) $x + y'$ (d) $x' + y$

Computer – Boolean Expression Simplification – 2021

85. Which of the following logic operations is performed by the following given combinational circuit?



- (a) Exclusive-OR (b) Exclusive-NOR
(c) NAND (d) NOR

Computer – Logic Gates – 2021

86. The output of an AND gate with three inputs A, B and C is HIGH when

- (a) $A = 1, B = 1, C = 0$ (b) $A = 0, B = 0, C = 0$
(c) $A = 1, B = 1, C = 1$ (d) $A = 1, B = 0, C = 1$

Computer – AND Gate – 2021

87. 2's complement of 1011011 is

- (a) 0100011 (b) 0110101
(c) 0100111 (d) 0100101

Computer – 2's Complement – 2021

88. A digital circuit that can store only one bit is a/an

- (a) register (b) NOR gate
(c) flip-flop (d) XOR gate

Computer – Flip-Flops – 2021

89. How much input and output needed for demultiplexer?

- (a) Many inputs one output
(b) One input many outputs
(c) One input one output
(d) None of the above

Computer – Demultiplexer – 2021

90. If A, B and C are the inputs of a full adder, then the sum is given by

- (a) A and B And C (b) A OR B And C
(c) A XOR B XOR C (d) A OR B OR C

Computer – Full Adder – 2021

91. The decimal equivalent of the binary number $(1011.01)_2$

- (a) $(11.375)_{10}$ (b) $(10.123)_{10}$
(c) $(11.25)_{10}$ (d) $(9.23)_{10}$

Computer – Binary to Decimal Conversion – 2021

92. Don't care conditions can be used for simplifying Boolean expressions in

- (a) registers (b) terms
(c) K-maps (d) latches

Computer – Karnaugh Map – 2021

93. A D flip-flop can be constructed from a/an ____ flip-flop.

- (a) S-R (b) J-K
(c) T (d) S-K

Computer – Flip-Flop Conversion – 2021

94. How many full adder and half adder are required to add 16-bit numbers?

- (a) 8 half adders, 8 full adders
(b) 1 half adder, 15 full adders
(c) 16 half adders, no full adder
(d) 4 half adders, 12 full adders

Computer – Adders – 2021

95. Which flip-flops serves to be the fundamental building blocks of counters?

- (a) S-R flip-flops (b) J-K flip-flops
(c) T flip-flops (d) D flip-flops

Computer – Counters – 2021

96. What is the difference between a shift-right register and a shift-left register?

- (a) There is no difference
(b) The direction of the shift
(c) Propagation delay
(d) The clock input

Computer – Shift Registers – 2021

97. Which is not characteristic of a shift register?

- (a) Serial in/parallel in (b) Serial in/ parallel out
(c) Parallel in / serial out (d) Parallel in / Parallel out

Computer – Shift Register Types – 2021





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98. In the expression $A + BC$, the total numbers of minterms will be

- (a) 2 (b) 3
- (c) 4 (d) 5

Computer – Boolean Minterms – 2021

99. Which of the following is a structured programming technique that graphically represents the detailed steps required to solve a program?

- (a) Algorithm (b) Pseudocode
- (c) Flowchart (d) Top-down design

Computer – Flowchart – 2021

100. Which of the following is a group of bits that tells the computer to perform a particular operation?

- (a) Accumulator (b) Register
- (c) Instruction code (d) None of these

Computer – Instruction Code – 2021

ANSWER KEY

1	d	2	a	3	a	4	b	5	d
6	a	7	c	8	b	9	c	10	b
11	d	12	b	13	d	14	b	15	c
16	a	17	a	18	b	19	b	20	d
21	a	22	a	23	b	24	b	25	a
26	c	27	a	28	b	29	b	30	d
31	c	32	d	33	a	34	b	35	d
36	c	37	b	38	c	39	c	40	b
41	b	42	c	43	c	44	a	45	a
46	b	47	a	48	b	49	a	50	a
51	b	52	c	53	a	54	c	55	a
56	b	57	d	58	b	59	a	60	b
61	a	62	b	63	a	64	a	65	b
66	a	67	b	68	d	69	b	70	b
71	d	72	c	73	b	74	d	75	a
76	c	77	b	78	b	79	b	80	b
81	b	82	a	83	c	84	d	85	a
86	c	87	d	88	c	89	b	90	c
91	c	92	c	93	a	94	b	95	c
96	b	97	a	98	d	99	c	100	c

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9554548576



ENGLISH SOLUTION

11. (d) 28

12.

$$40 \times 12 + 3 - 6 + 60 = ?$$

$$40 + 12 \div 3 \times 60$$

$$39798 + 3798 + 378 = ?$$

$$43974$$

15. Lowest common multiple
24, 36, 40
= 360

16. Option (a) 14

$$15 \times 14 = 14 + 196$$

$$210 \quad 210$$

Ans. 14

18. Sword

19. 21 = 3×7
49 = 7×7
36 = 8×7
= 36 Ans.

20. 1, 3, 3, 6, 7, 9, ?, 12, 2

$$\begin{array}{ccccccc} & +2 & & +4 & & +6 & & +8 \\ & \downarrow & & \downarrow & & \downarrow & & \downarrow \end{array}$$

= 13 Ans.

21.	T	W	E	N	T	Y
	8	6	3	9	8	5
	E	L	E	V	E	N
	3	2	3	0	3	9
	T	W	E	L	V	E
	8	6	3	2	0	3

22. DGCA – Directorate General of Civil Aviation

23. Joe Biden

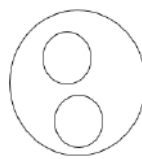
23. $5 + 3 + 3 + 4$

25. Sputnik V

26. {1, 2, 3} and {1, 2, 5}
{3}

27. Infinite

28.



29.

$n(A) = P$
 $n(B) = q$
How many ordered pair
(a)(b)(c)(d), b) are there
 $a \in A$
 $b \in B$
= $P \times q$ Ans.

30. $S = \{[\phi], 1, [2, 3]\}$
 $S = 3$ element
No. of element is power set
= $2^3 = 8$

31. Finite set

32. The Characteristics of the relation aRb if $a^2 = b^2$
= Symmetric, Reflexive and transitive

33. $n(E) = 18$
 $n(H) = 15$
 $n(K) = 22$
 $n(E \cap H) = 9$
 $n(H \cap K) = 11$
 $n(K \cap E) = 13$
 $n(K \cap E \cap H) = ?$
 $n(K \cup E \cup H) = 28$
 $n(K \cup E \cup H) = n(E) + n(H) +$
 $n(K) - (n(E \cap H) + n(H \cap K) +$
 $+ n(K \cap E) + n(K \cap H \cap E))$

$$28 = 18 + 15 + 22 - (9 + 11 + 13) + 2$$

$$28 = 55 - (33) + x$$

$$28 \pm 22 + x$$

$$\cap (K \cap E \cap H) = 6 \text{ Ans.}$$

34. Drop

35. Billman - Ford
Shortest path algorithm

37. Vertices and edges.

38. Gaussian distribution



39. BANANA

$$\frac{6!}{3! \times 2!} = \frac{6 \times 5 \times 4}{2} = 60$$

40. $P(A) = \frac{x}{3}$

$$P(\bar{A}) = \frac{2}{3}$$

$$1 - \frac{x}{3} = \frac{2}{3}$$

$$3 - x = 2$$

$$3 - 2 = x$$

$$X = 1 \text{ Ans.}$$

41. x, y, z

$$SD = P$$

Then the SD of

$$3x + 5, 3y + 5, 3z + 5$$

$$X, y, z \Rightarrow SD = 3P$$

$$3x + 5, 3y + 5, 3z + 5$$

↓

$$SD = 3P \text{ Ans.}$$

42. The man has selected some or all of his 6 friends. So required number of wags.

$$= 2^6 - 1 = 63$$

43. $a_n = 2a_{n-1} + 3$

$$a_0 = 6$$

$$a_1 = 2a_{1-1} + 3$$

$$a_1 = 2a_0 + 3 = 15$$

$$a_2 = 2a_1 + 3 = 33$$

$$a_3 = 2a_2 + 3 = 69$$

$$a_4 = 2a_3 + 3 = 141$$

$$a_4 = 141 \text{ Days.}$$

44. Shortest job first.

45. $A^2 = A$

Idempotent Matrix

$$P(K+1) = m^{(K+1)} + 5 \text{ is true}$$

Then

$$= P(K) = m^K + 5 \text{ Ans.}$$

47. Mesh → Ans.

48. $A = \begin{bmatrix} a & b \\ b & a \end{bmatrix}$ and

$$A^2 \begin{bmatrix} \alpha & \beta \\ \beta & \alpha \end{bmatrix}$$

$$A^2 \begin{bmatrix} a & b \\ b & a \end{bmatrix} \begin{bmatrix} a & b \\ b & a \end{bmatrix}$$

$$= \begin{bmatrix} a^2 + b^2 & ab^2 + ab \\ ab + ab & b^2 + a^2 \end{bmatrix}$$

$$= \begin{bmatrix} a^2 + b^2 & 2ab \\ 2ab & a^2 + b^2 \end{bmatrix}$$

$$\alpha = a^2 + b^2$$

$$\beta = 2ab \text{ Ans.}$$

49. A few long CPU bursts

51. Delete an Element

52. Pseudo Code

53. One stack is enough.

54. new

55. O(n)

56. 1,1

57. -128

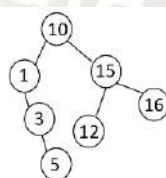
59. it is the ability for a message/data to processed is more than one from.

60. Ada

62. Unicode

63. Postfix

64.



The maximum distance of a leaf nodes
From the root is 3.

65. Divide and conquer algorithm

66. Friend function

68. To search in unordered list

70. Greedy algorithm

71. All of the above.

72. Search Key value = 8 bytes

Block size = 512 byte

Block pointer = 2 bytes

$$\text{Max key per block} = \frac{(\text{Block size} - \text{Block Pointer})}{\text{Search Key size}}$$

$$= \frac{512-2}{8}$$

$$= 63$$

Max order = (Max Keyper)



Block +1

$$\frac{63+1}{2} = 32 \text{ Ans.}$$

Therefore the maximum number of the B true is 52 Ans.

73. 1 Function

74. 2

75. Copy of re.... is created.

76. 132

77. Only inside the {} block

78. Counting the number of key operation

81. $O \log(n)$

82. $Y + x^1 \cdot Y$

= ?

Option (a)

83. $x \cdot x' = 1$

84. $x' y' + xy + x'y$

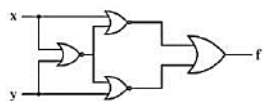
$$xy + x' (y' + y)$$

$$xy + x'$$

$$(x + x') (y + y')$$

$$y + x' \text{ Ans.}$$

85. $(x + 1 x + y)'$



$$x' y + xy'$$

↓

Exclusive – OR

Ans. = Exclusive OR

86. $A = 1, B = 1, C = 1$

87. 2's complement of

$$1011011$$

↓

$$0100101 \text{ Ans.}$$

88. Flip Flop

90. A XOR B XOR C

91.

$$\begin{matrix} (1) & 0 & 1 & 1 & 0 & 1 \\ \downarrow & \downarrow & \downarrow & \downarrow & \downarrow & \downarrow \\ 2^3 & 2^2 & 2^1 & 2^0 & 2^{-1} & 2^{-2} \end{matrix}$$

$$8 + 4 + 1 = 13$$

$$(11.375)_{10}$$

92. K-Maps

93. S.R Flipflop

97. Serial is /Parallel is

$$98. Y = A + BC$$

$$Y = A(B + \bar{B}) + (A + \bar{A})BC$$

$$Y = AB(C + \bar{C}) + A\bar{B}\bar{C} + AB$$

$$Y = \bar{A}BC + ABC + ABC + A\bar{B}\bar{C} + A\bar{B}C + AB\bar{C}$$

$$Y = \bar{A}BC + ABC + A\bar{B}\bar{C} + A\bar{B}C + AB\bar{C}$$

$$= 5 \text{ min term Ans.}$$

99. Flowchart

100. Instruction Code